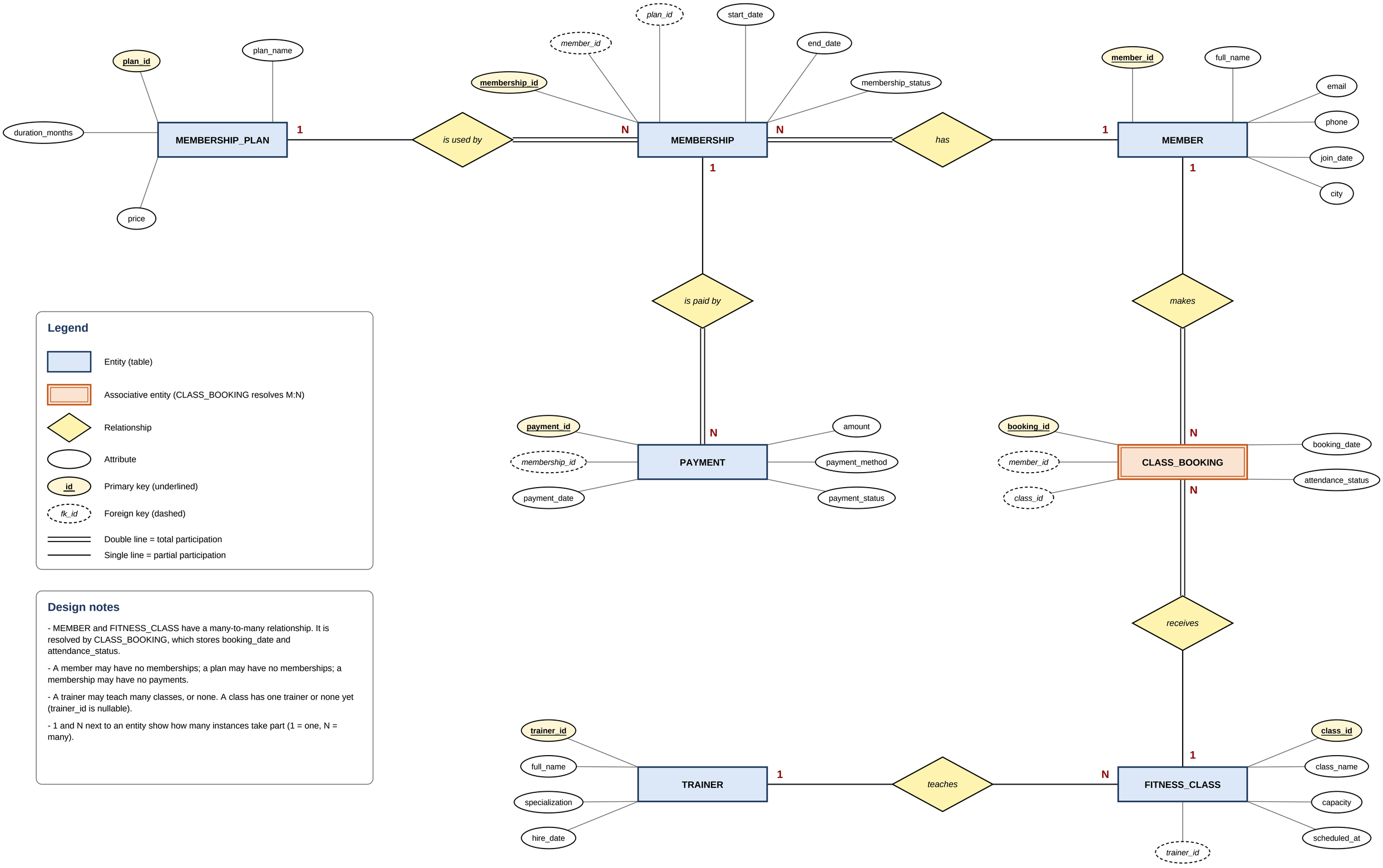
Gym Membership and Management System

Entity Relationship Diagram (ERD) - Chen Notation



Legend

- Entity (table)
- Associative entity (CLASS_BOOKING resolves M:N)
- Relationship
- Attribute
- **id** — Primary key (underlined)
- *fk_id* — Foreign key (dashed)
- Double line = total participation
- Single line = partial participation

Design notes

- MEMBER and FITNESS_CLASS have a many-to-many relationship. It is resolved by CLASS_BOOKING, which stores booking_date and attendance_status.
- A member may have no memberships; a plan may have no memberships; a membership may have no payments.
- A trainer may teach many classes, or none. A class has one trainer or none yet (trainer_id is nullable).
- 1 and N next to an entity show how many instances take part (1 = one, N = many).

Relationship Summary

#	Entity A	Relationship	Entity B	Cardinality (A : B)	Foreign Key	Business rule
1	MEMBER	has	MEMBERSHIP	1 : N	MEMBERSHIP.member_id	A member may have many memberships over time, or none. Each membership belongs to exactly one member.
2	MEMBERSHIP_PLAN	is used by	MEMBERSHIP	1 : N	MEMBERSHIP.plan_id	A plan can be used by many memberships, or currently none. Each membership uses exactly one plan.
3	MEMBERSHIP	is paid by	PAYMENT	1 : N	PAYMENT.membership_id	A membership may have no payment or several payments. Each payment belongs to exactly one membership.
4	TRAINER	teaches	FITNESS_CLASS	1 : N	FITNESS_CLASS.trainer_id (nullable)	A trainer may teach many classes, or none. A class has one assigned trainer or no trainer yet.
5	MEMBER	makes	CLASS_BOOKING	1 : N	CLASS_BOOKING.member_id	A member can make many bookings, or none. Each booking is made by exactly one member.
6	FITNESS_CLASS	receives	CLASS_BOOKING	1 : N	CLASS_BOOKING.class_id	A class can receive many bookings, or none. Each booking is for exactly one class.
7	MEMBER	books (via CLASS_BOOKING)	FITNESS_CLASS	M : N	-	Many-to-many relationship resolved by CLASS_BOOKING, which also stores booking_date and attendance_status.